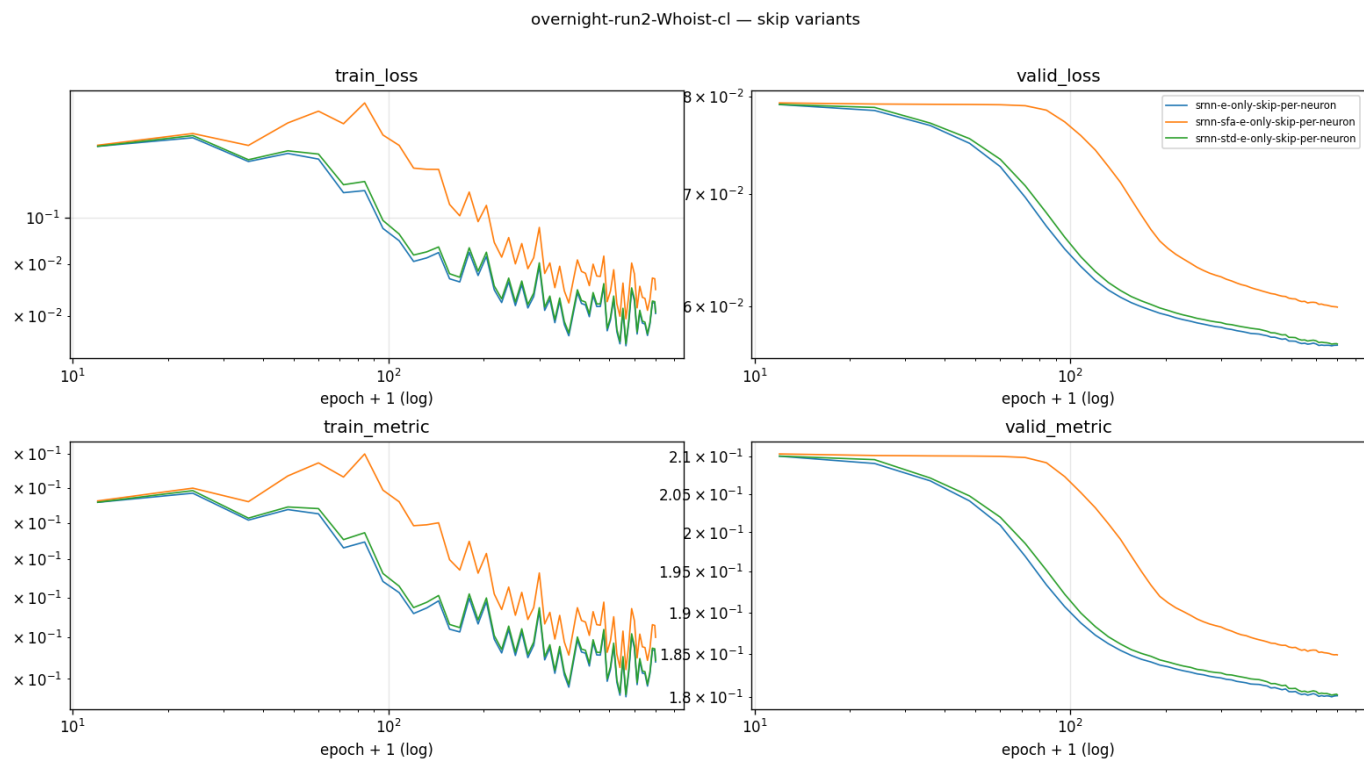


Run: overnight-run2-Whoist-cl

Log-log loss curves



Final test_loss (init - last) [skip]:
 srnn-e-only-skip-per-neuron: 0.07877 → 0.05486
 srnn-sfa-e-only-skip-per-neuron: 0.09420 → 0.05791
 srnn-std-e-only-skip-per-neuron: 0.07907 → 0.05505

Figure 1: skip variants — log-log loss/metric

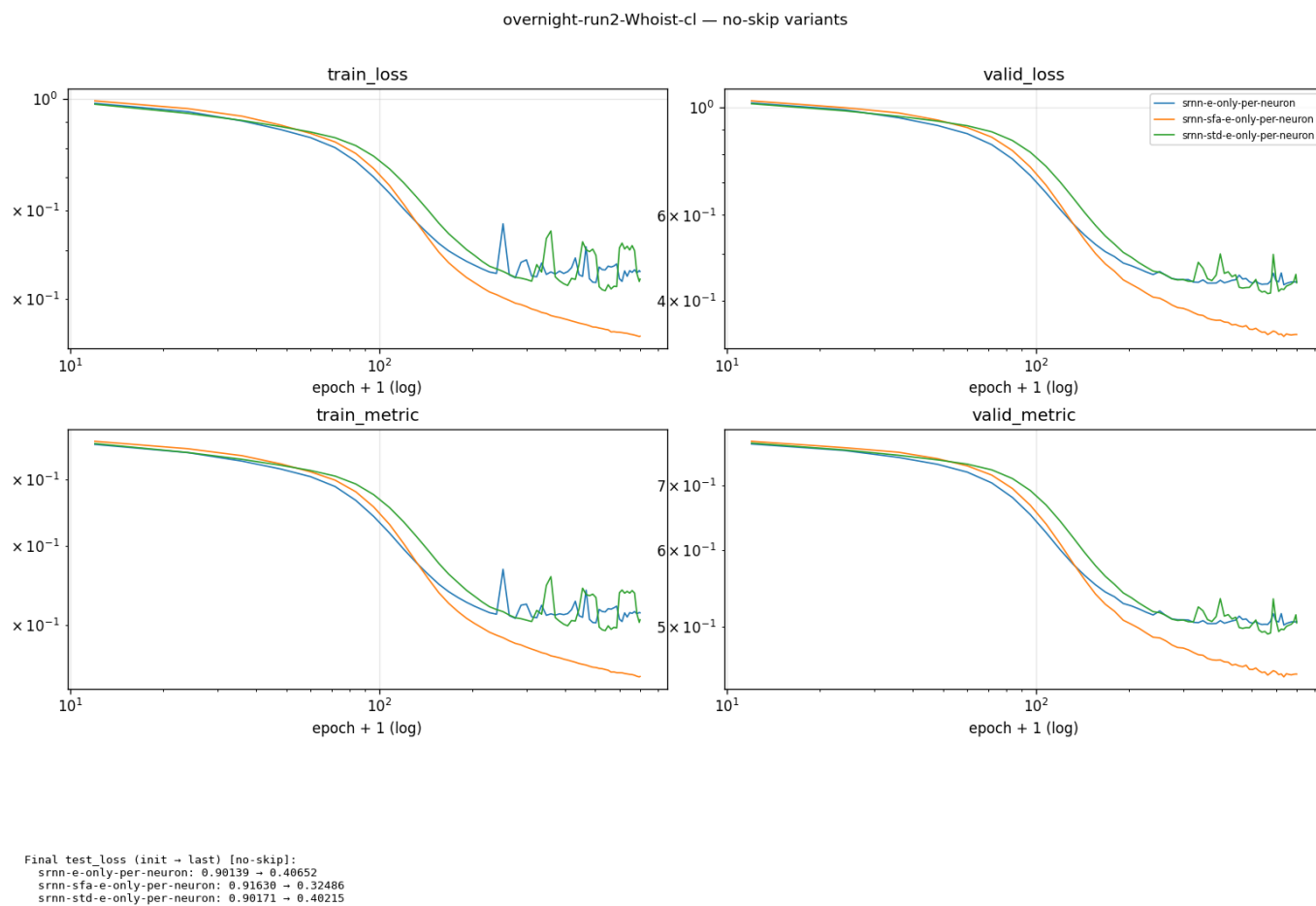


Figure 2: no-skip variants — log-log loss/metric

srnn-e-only-per-neuron

overnight-run2-Whoist-cl — effective time constants (srnn-e-only-per-neuron)

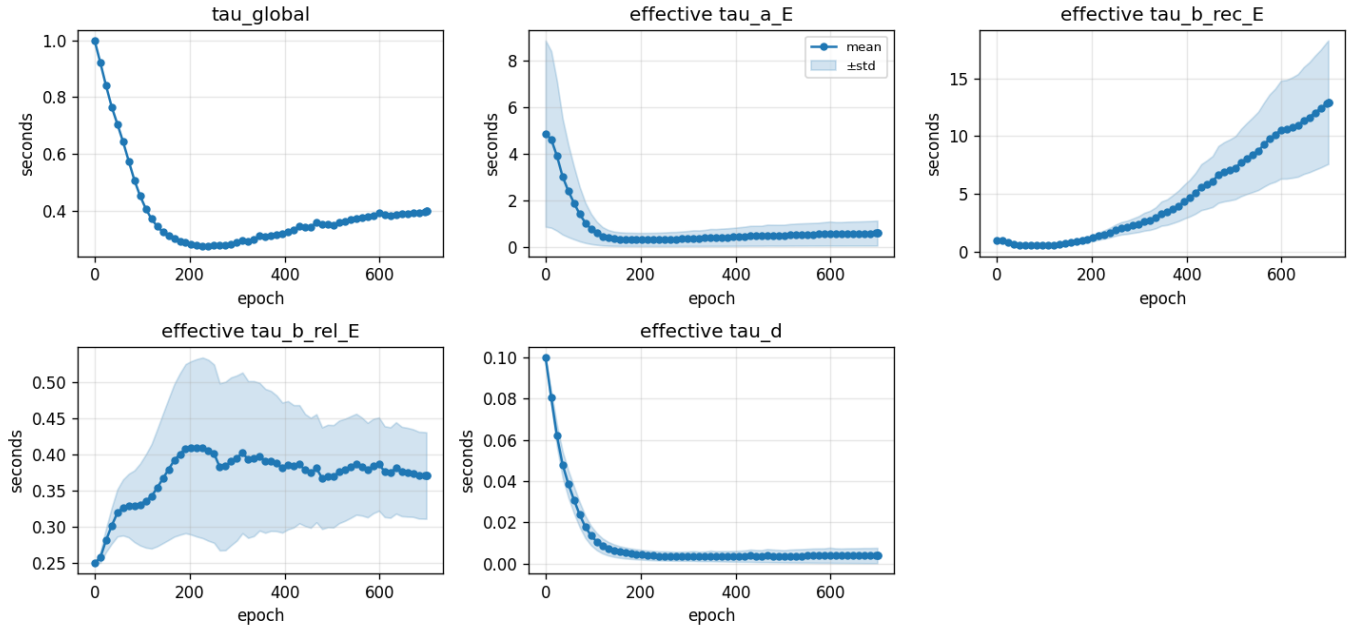
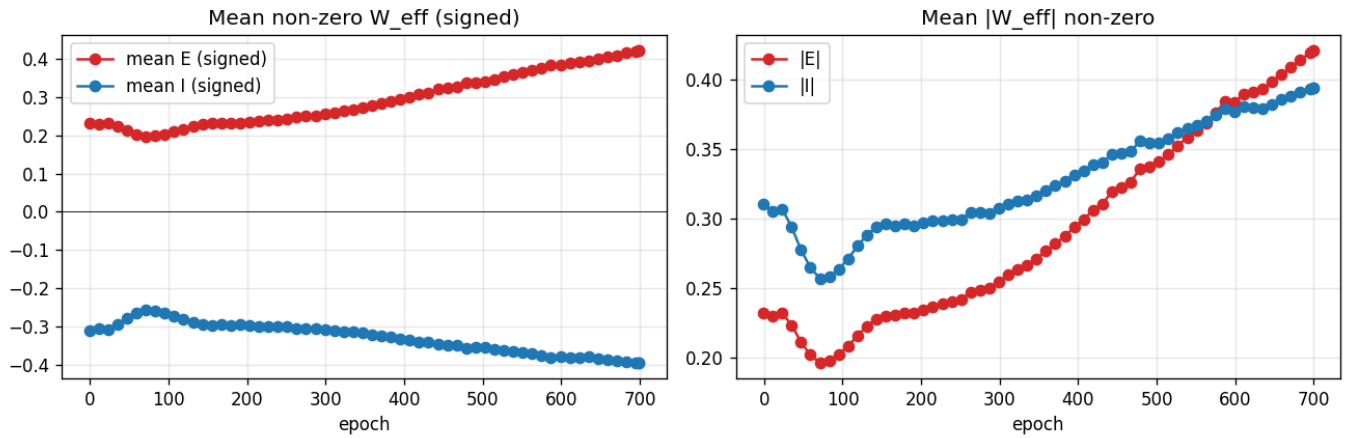


Figure 3: Tau evolution

overnight-run2-Whoist-cl — recurrent E vs I weights (srnn-e-only-per-neuron)

Figure 4: W_{EI} evolution

srnn-e-only-per-neuron — parameter table

```
# overnight-run2-Whoist-cl - variant srnn-e-only-per-neuron (k=0)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.39706	-	-	scalar
tau_d (s)	+0.1	+0.0039354	0	0.003722	300
tau_a_E (s)	+4.8777	+0.6065	3.996	0.5346	450
tau_b_rec_E (s)	+1	+12.939	0	5.352	150
tau_b_rel_E (s)	+0.25	+0.37073	0	0.05959	150
W_eff (E src, signed)	+0.23267	+0.42072	0.07641	0.363	15048
W_eff (I src, signed)	-0.31062	-0.39413	0.07724	0.1694	14977
W_eff (E src)	+0.23267	+0.42072	0.07641	0.363	15048
W_eff (I src)	+0.31062	+0.39413	0.07724	0.1694	14977
W_in_eff (input neurons)	-0.0003386	+0.00014097	0.1012	0.07117	6675
c_E (SFA coupling)	+0.05	+0.34718	0	0.4698	450
c_0_E (SFA offset)	+0	-1.8195	0	0.6839	450
a_0 (threshold)	+0.35	-1.1487	0	0.2353	300
readout_weight	-2.8854e-05	-0.01866	0.06604	1.029	6675
readout_bias	+0	+0.030146	0	0.1201	89
ic.ic (per-variant init)	+0.10386	+0.10386	0.3828	0.3828	900

srnn-e-only-skip-per-neuron

overnight-run2-Whoist-cl — effective time constants (srnn-e-only-skip-per-neuron)

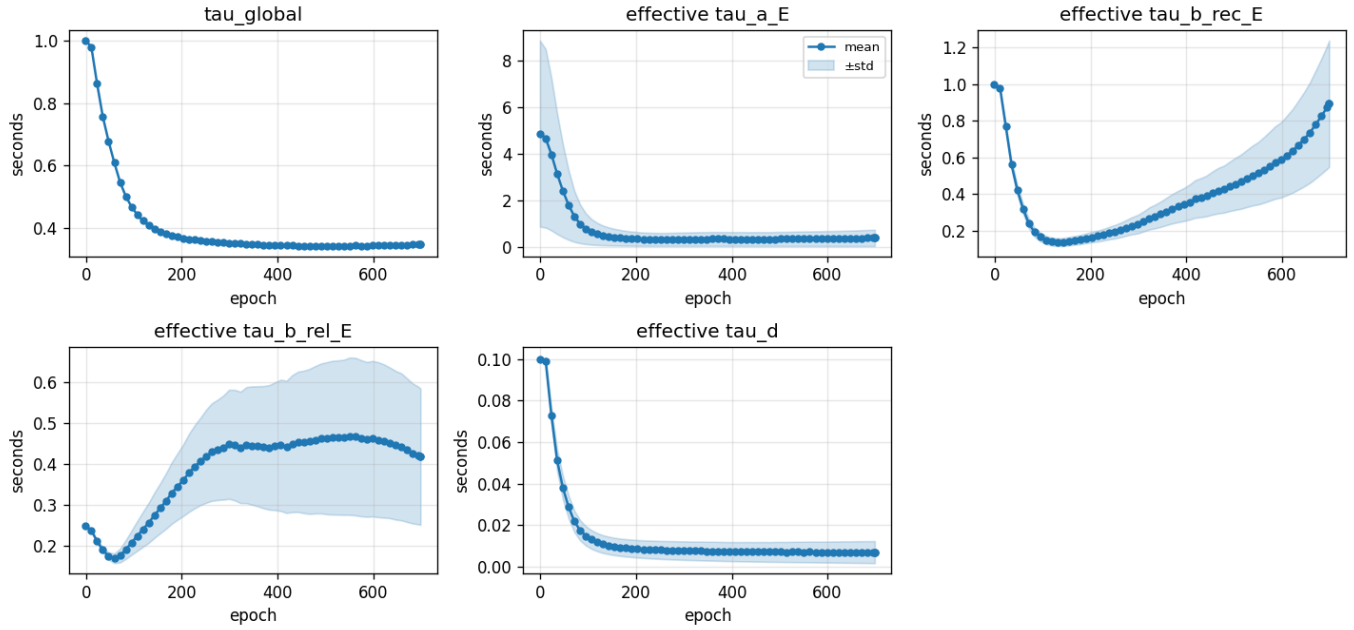
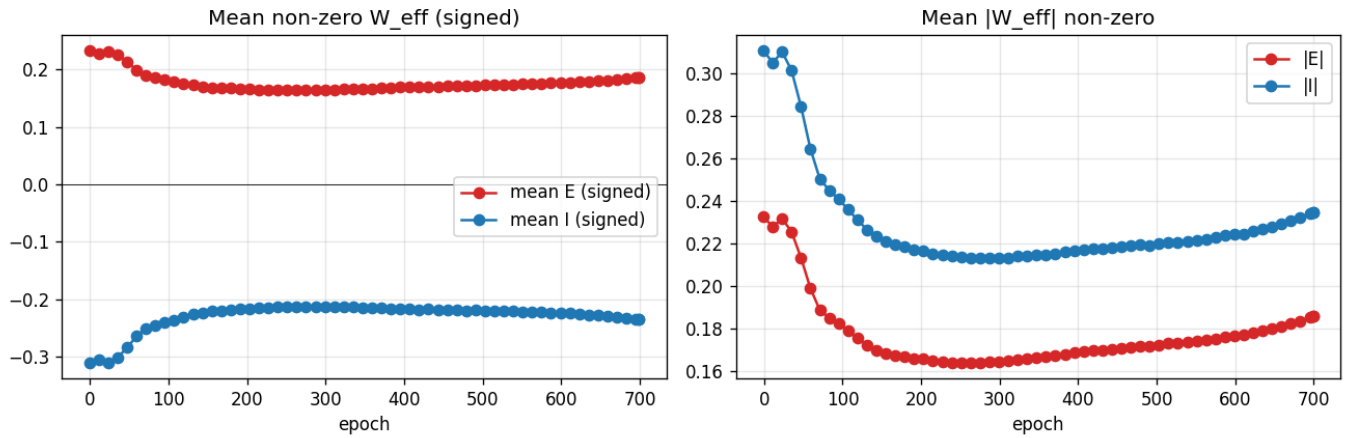


Figure 5: Tau evolution

overnight-run2-Whoist-cl — recurrent E vs I weights (srnn-e-only-skip-per-neuron)

Figure 6: W_{EI} evolution

srnn-e-only-skip-per-neuron — parameter table

```
# overnight-run2-Whoist-cl - variant srnn-e-only-skip-per-neuron (k=1)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.34822	-	-	scalar
tau_d (s)	+0.1	+0.0070211	0	0.005324	300
tau_a_E (s)	+4.8777	+0.39992	3.996	0.3518	450
tau_b_rec_E (s)	+1	+0.89423	0	0.3454	150
tau_b_rel_E (s)	+0.25	+0.41824	0	0.1665	150
W_eff (E src, signed)	+0.23267	+0.18583	0.07641	0.09793	15048
W_eff (I src, signed)	-0.31062	-0.2346	0.07724	0.1038	14977
W_eff (E src)	+0.23267	+0.18583	0.07641	0.09793	15048
W_eff (I src)	+0.31062	+0.2346	0.07724	0.1038	14977
W_in_eff (input neurons)	-0.00084237	-0.00032963	0.1008	0.03327	6675
c_E (SFA coupling)	+0.05	+0.16025	0	0.1934	450
c_0_E (SFA offset)	+0	-0.51227	0	0.2784	450
a_0 (threshold)	+0.35	-0.45279	0	0.1396	300
readout_weight	+0.00078219	-0.0049355	0.06654	0.2849	6675
readout_bias	+0	+0.14971	0	0.2096	89
ic.ic (per-variant init)	+0.10327	+0.10327	0.3829	0.3829	900

srnn-sfa-e-only-per-neuron

overnight-run2-Whoist-cl — effective time constants (srnn-sfa-e-only-per-neuron)

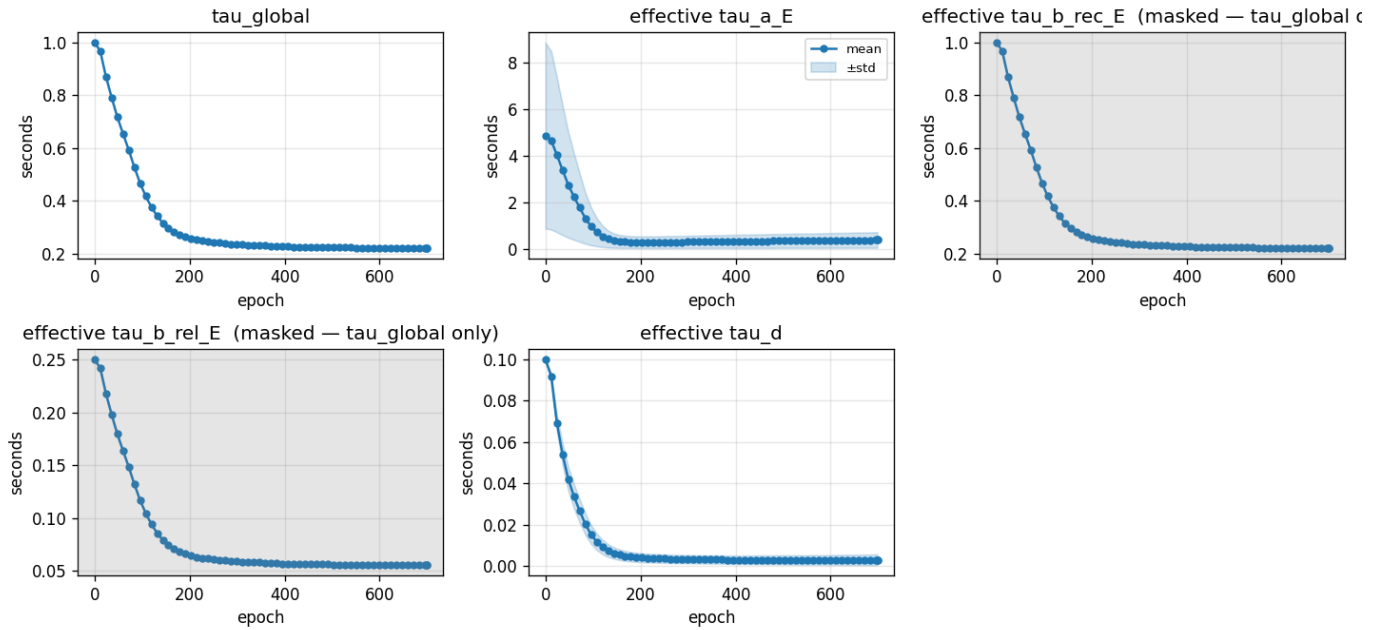
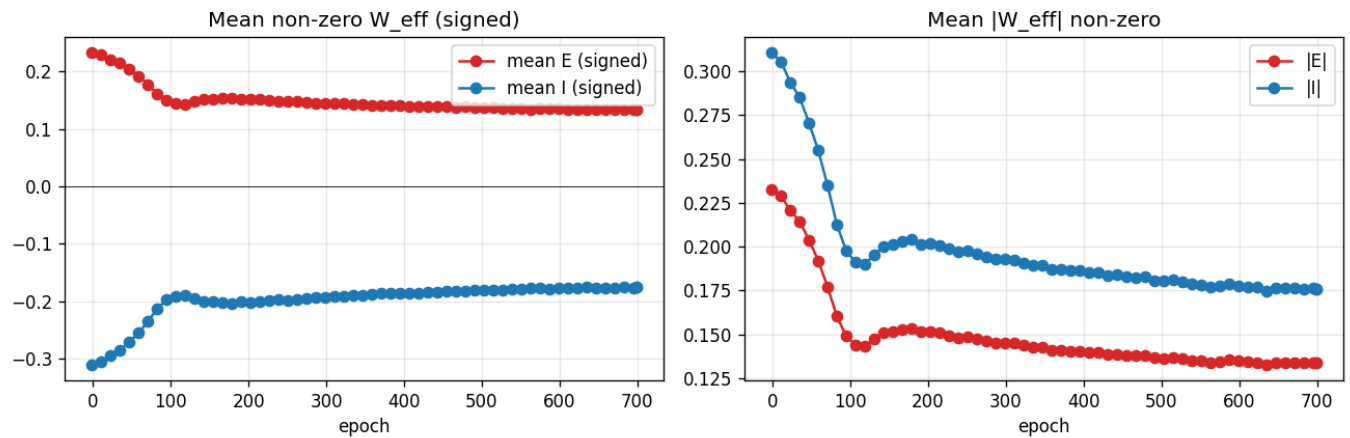


Figure 7: Tau evolution

overnight-run2-Whoist-cl — recurrent E vs I weights (srnn-sfa-e-only-per-neuron)

Figure 8: W_{EI} evolution

srnn-sfa-e-only-per-neuron — parameter table

```
# overnight-run2-Whoist-cl - variant srnn-sfa-e-only-per-neuron (k=2)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.22226	-	-	scalar
tau_d (s)	+0.1	+0.0029146	0	0.002612	300
tau_a_E (s)	+4.8777	+0.39283	3.996	0.3336	450
tau_b_rec_E (s)	+1	+0.22226	0	0	150
tau_b_rel_E (s)	+0.25	+0.055565	0	0	150
W_eff (E src, signed)	+0.23267	+0.13375	0.07641	0.06408	15048
W_eff (I src, signed)	-0.31062	-0.17607	0.07724	0.0687	14977
W_eff (E src)	+0.23267	+0.13375	0.07641	0.06408	15048
W_eff (I src)	+0.31062	+0.17607	0.07724	0.0687	14977
W_in_eff (input neurons)	-0.00015382	-0.00086168	0.09934	0.06373	6675
c_E (SFA coupling)	+0.05	+0.35569	0	0.4511	450
c_0_E (SFA offset)	+0	-0.25393	0	0.26	450
a_0 (threshold)	+0.35	+0.0098035	0	0.131	300
readout_weight	-0.00098972	+0.0069969	0.06678	0.4859	6675
readout_bias	+0	-0.13424	0	0.2878	89
ic.ic (per-variant init)	+0.13078	+0.13078	0.4172	0.4172	900

srnn-sfa-e-only-skip-per-neuron

overnight-run2-Whoist-cl — effective time constants (srnn-sfa-e-only-skip-per-neuron)

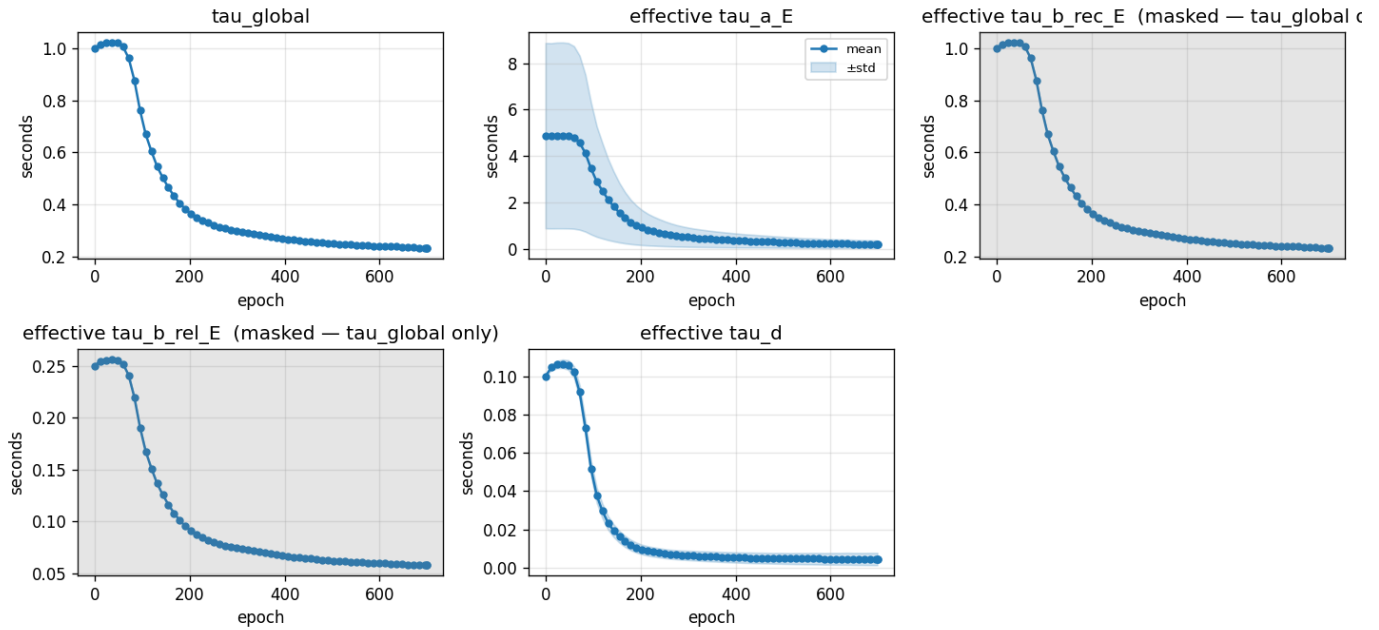
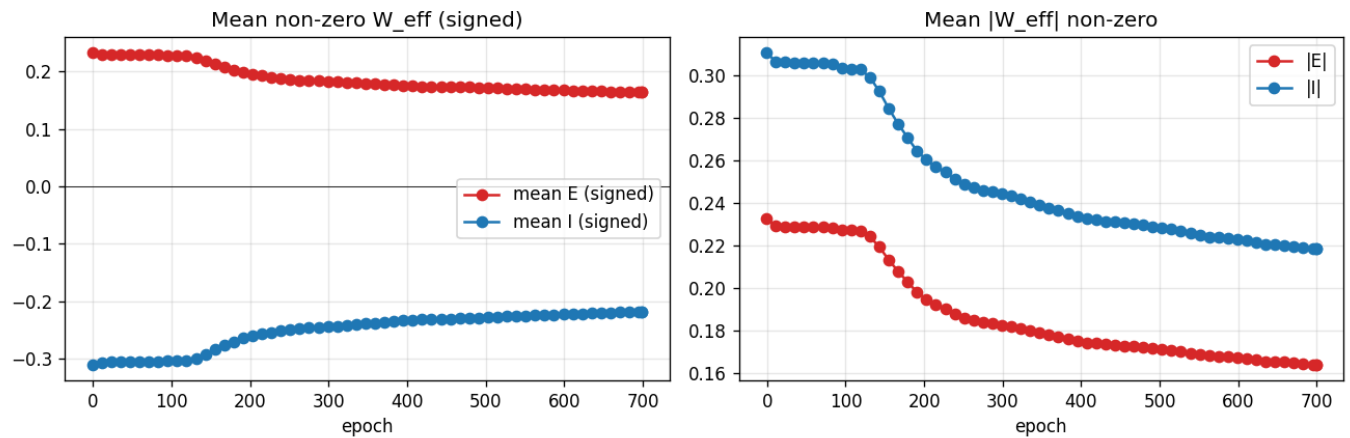


Figure 9: Tau evolution

overnight-run2-Whoist-cl — recurrent E vs I weights (srnn-sfa-e-only-skip-per-neuron)

Figure 10: W_{EI} evolution

srnn-sfa-e-only-skip-per-neuron — parameter table

```
# overnight-run2-Whoist-cl - variant srnn-sfa-e-only-skip-per-neuron (k=3)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.23219	-	-	scalar
tau_d (s)	+0.1	+0.0045342	0	0.003217	300
tau_a_E (s)	+4.8777	+0.19658	3.996	0.1663	450
tau_b_rec_E (s)	+1	+0.23219	0	0	150
tau_b_rel_E (s)	+0.25	+0.058049	0	0	150
W_eff (E src, signed)	+0.23267	+0.16382	0.07641	0.06763	15048
W_eff (I src, signed)	-0.31062	-0.2183	0.07724	0.07632	14977
W_eff (E src)	+0.23267	+0.16382	0.07641	0.06763	15048
W_eff (I src)	+0.31062	+0.2183	0.07724	0.07632	14977
W_in_eff (input neurons)	-0.0011621	+9.8282e-05	0.09962	0.03324	6675
c_E (SFA coupling)	+0.05	+0.068387	0	0.04854	450
c_0_E (SFA offset)	+0	-0.21228	0	0.2366	450
a_0 (threshold)	+0.35	+0.18645	0	0.1337	300
readout_weight	+0.00016801	-0.010457	0.0673	0.2114	6675
readout_bias	+0	+0.10138	0	0.2415	89
ic.ic (per-variant init)	+0.09231	+0.09231	0.5294	0.5294	900

srnn-std-e-only-per-neuron

overnight-run2-Whoist-cl — effective time constants (srnn-std-e-only-per-neuron)

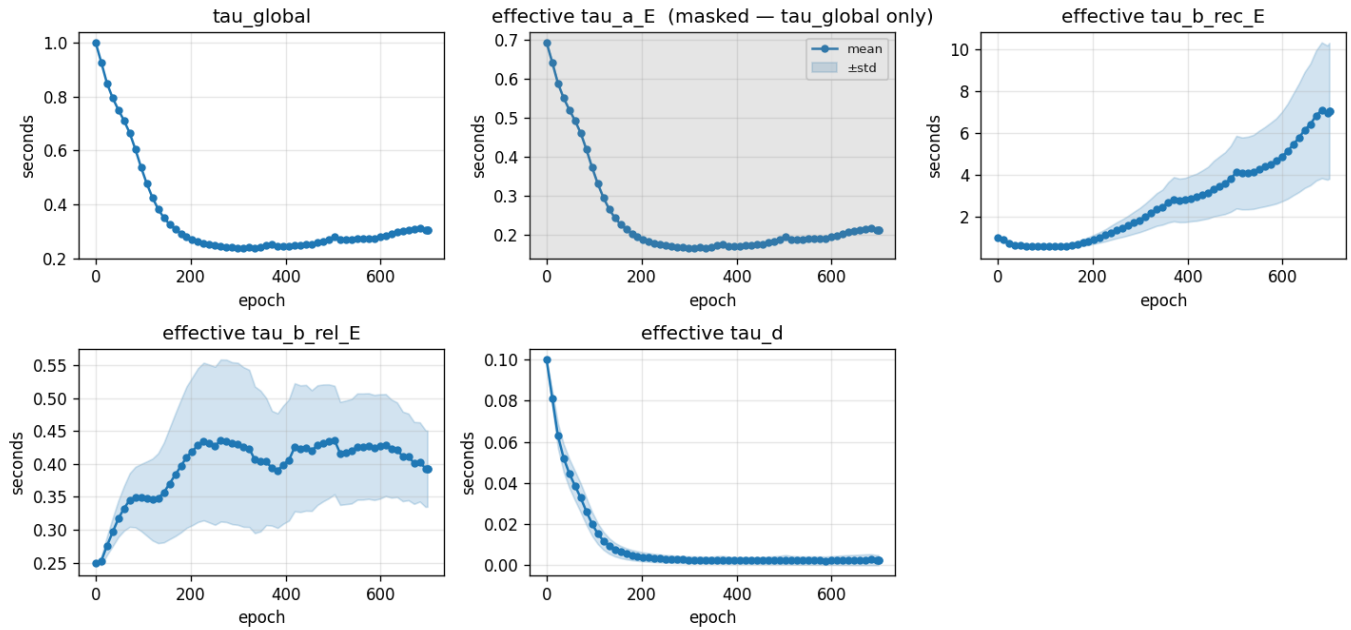
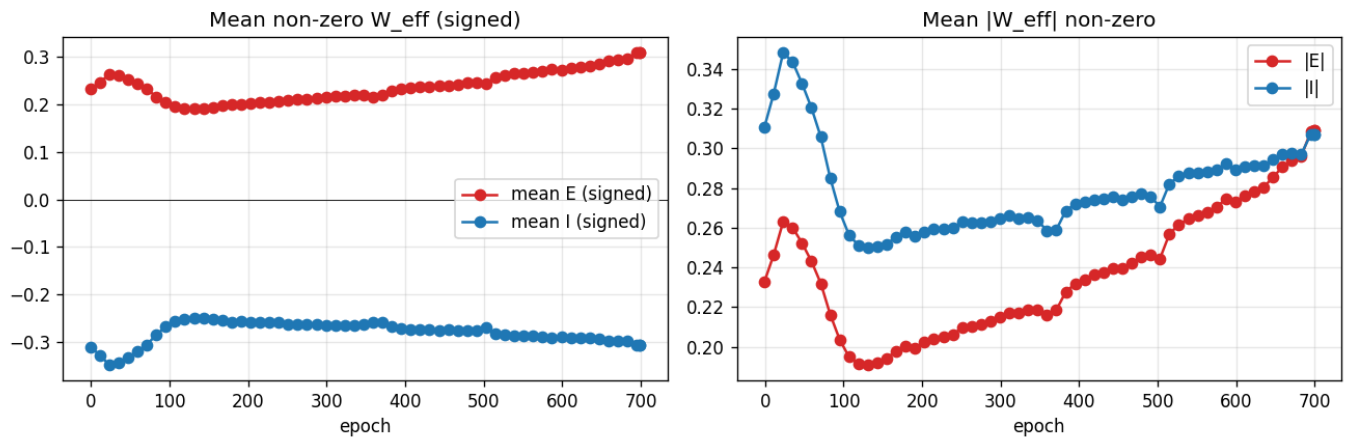


Figure 11: Tau evolution

overnight-run2-Whoist-cl — recurrent E vs I weights (srnn-std-e-only-per-neuron)

Figure 12: W_{EI} evolution

srnn-std-e-only-per-neuron — parameter table

```
# overnight-run2-Whoist-cl - variant srnn-std-e-only-per-neuron (k=4)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.30412	-	-	scalar
tau_d (s)	+0.1	+0.0024888	0	0.002472	300
tau_a_E (s)	+0.69315	+0.2108	0	0	450
tau_b_rec_E (s)	+1	+7.0621	0	3.241	150
tau_b_rel_E (s)	+0.25	+0.39294	0	0.0579	150
W_eff (E src, signed)	+0.23267	+0.30903	0.07641	0.2289	15048
W_eff (I src, signed)	-0.31062	-0.30695	0.07724	0.126	14977
W_eff (E src)	+0.23267	+0.30903	0.07641	0.2289	15048
W_eff (I src)	+0.31062	+0.30695	0.07724	0.126	14977
W_in_eff (input neurons)	+0.00019944	-0.00042256	0.09994	0.05212	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	-1.6603	0	0.2753	300
readout_weight	+0.00074661	-0.0033378	0.06657	0.9113	6675
readout_bias	+0	+0.013818	0	0.1603	89
ic.ic (per-variant init)	+0.090819	+0.090819	0.3835	0.3835	900

srnn-std-e-only-skip-per-neuron

overnight-run2-Whoist-cl — effective time constants (srnn-std-e-only-skip-per-neuron)

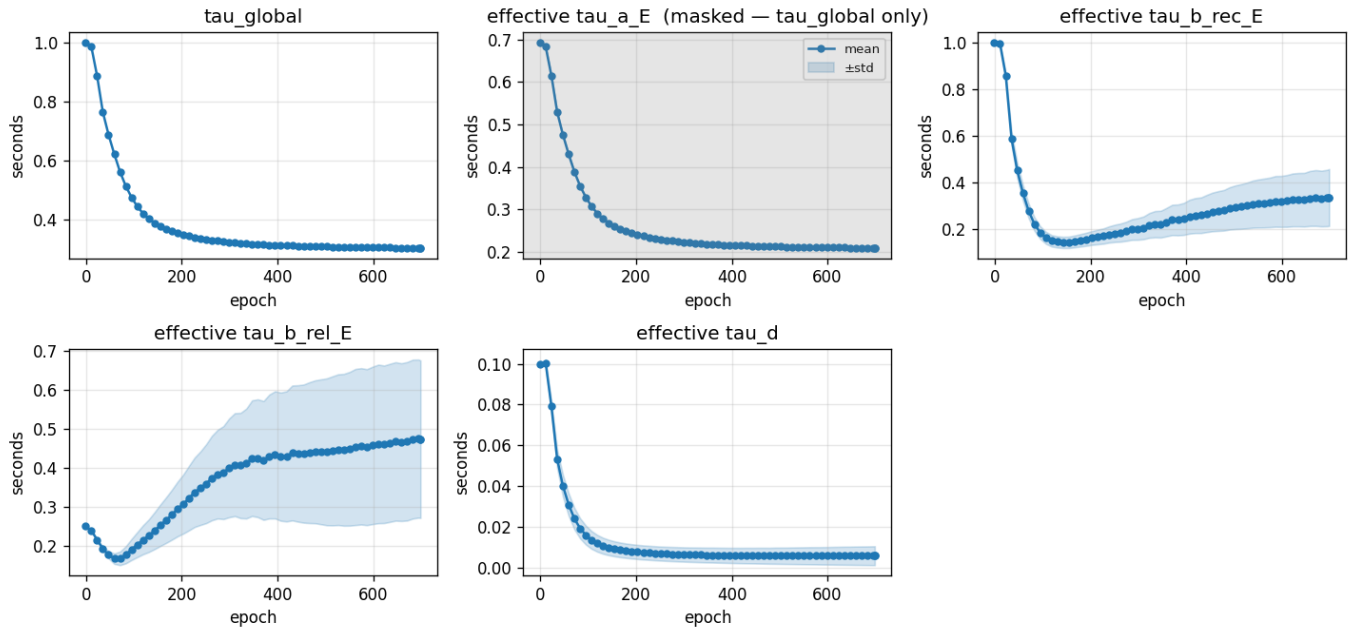
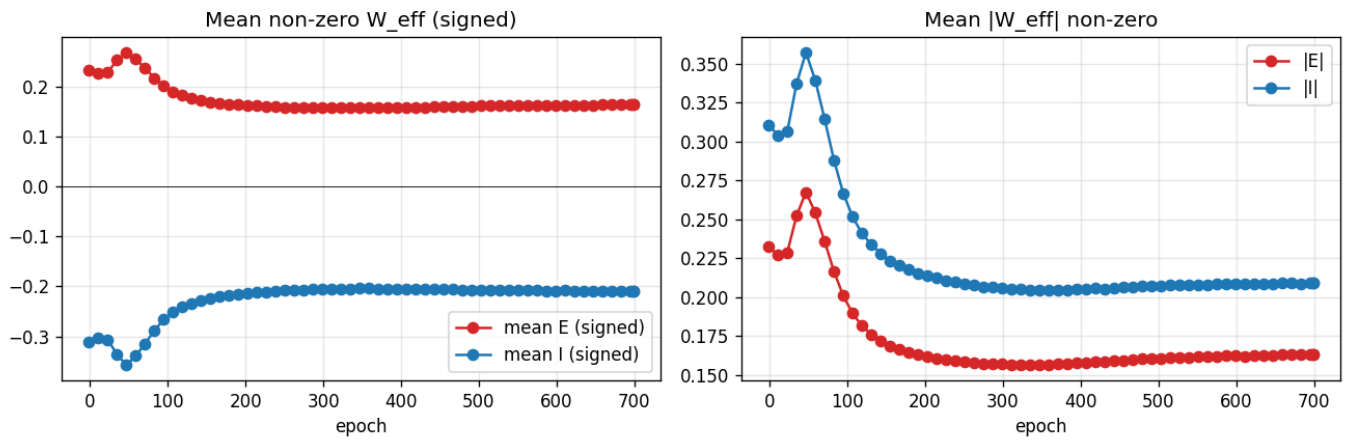


Figure 13: Tau evolution

overnight-run2-Whoist-cl — recurrent E vs I weights (srnn-std-e-only-skip-per-neuron)

Figure 14: W_{EI} evolution

srnn-std-e-only-skip-per-neuron — parameter table

```
# overnight-run2-Whoist-cl - variant srnn-std-e-only-skip-per-neuron (k=5)
# Effective values (post-transform). Means/stds computed over the indicated dimension.
```

effective parameter	init mean	final mean	init std	final std	n

tau_global (s)	+1	+0.30196	-	-	scalar
tau_d (s)	+0.1	+0.0057643	0	0.004652	300
tau_a_E (s)	+0.69315	+0.2093	0	0	450
tau_b_rec_E (s)	+1	+0.33406	0	0.1223	150
tau_b_rel_E (s)	+0.25	+0.47352	0	0.2022	150
W_eff (E src, signed)	+0.23267	+0.16328	0.07641	0.08844	15048
W_eff (I src, signed)	-0.31062	-0.20913	0.07724	0.0967	14977
W_eff (E src)	+0.23267	+0.16328	0.07641	0.08844	15048
W_eff (I src)	+0.31062	+0.20913	0.07724	0.0967	14977
W_in_eff (input neurons)	+0.00036119	+0.00022384	0.09956	0.02544	6675
c_E (SFA coupling)	+0.05	+0.05	0	0	450
c_0_E (SFA offset)	+0	+0	0	0	450
a_0 (threshold)	+0.35	-0.33197	0	0.1515	300
readout_weight	+0.00055372	-0.00094974	0.06572	0.279	6675
readout_bias	+0	-0.034855	0	0.1837	89
ic.ic (per-variant init)	+0.083543	+0.083543	0.384	0.384	900